ch3 Change of State

Q1

Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Cappuccino is an Italian style coffee topped with a layer of frothy milk (Figure 1.1).

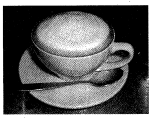

Figure 1.1

Frothy milk is made by bubbling steam through milk, which is held in a metallic jug (Figure 1.2). Steam is ejected from the steam wand of a cappuccino machine (Figure 1.3).

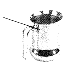

Figure 1.2

Figure 1.3

Given: specific latent heat of vaporization of water = 2.26×10^6 J kg^{-1}
specific heat capacity of water = 4200 J kg^{-1} °C^{-1}
specific heat capacity of steam = 2000 J kg^{-1} °C^{-1}
specific heat capacity of milk = 3900 J kg^{-1} °C^{-1}

(a) Calculate the total amount of heat released when 20 g of steam at 110°C cools to 100°C and condenses to water at 100°C. (3 marks)

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(b) 20 g of steam at 110°C is bubbled through 200 g of milk at 15°C to make frothy milk. Using the result in (a), estimate the temperature of the frothy milk. (2 marks)

(c) Would the actual temperature of frothy milk be higher than, equal to or lower than the value found in (b) ? Explain. (2 marks)

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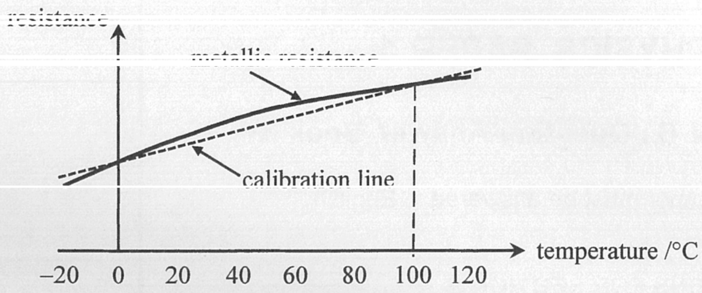
Q1

1. The solid curve in Figure 1.1 shows how the resistance of a metallic resistance thermometer varies with temperature. This thermometer is calibrated at standard atmospheric pressure for the melting point of ice and the steam point of boiling water. The dotted calibration line represents how the resistance of the thermometer varies with temperature if a linear resistance-temperature relationship is assumed. The deviation of the curve from linearity has been exaggerated in the figure.

Figure 1.1



(a) (i) Using the resistances at the calibration points tabulated below, calculate the expected resistance at 60 °C if the resistance varies linearly with temperature. (2 marks)

temperature /°C	resistance /Ω
0	102.00
100	140.51

(ii) Now if the resistance of the resistance thermometer is the value found in (a)(i), is the actual temperature higher than, lower than or equal to 60 °C ? (1 mark)

(b) In an experiment to determine the specific heat capacity of water c_W, Peter used this calibrated resistance thermometer to measure the temperature of water being heated from 0 °C to 60 °C. Heating was stopped when this thermometer's resistance reached the value found in (a)(i). Assuming negligible heat exchange with the surroundings, no error in measuring the energy supplied and the mass of water, explain whether the experimental value of c_W found is higher than, lower than or the same as the actual value. (2 marks)

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Q1

1. (a) An insulated container of negligible heat capacity contains 1.5 kg of tea at a temperature of 60 °C.

(i) What mass of ice at 0 °C should be added to the tea so that the final temperature of the mixture is lowered to 10 °C ? Assume that the specific heat capacity of tea is the same as that of water. (3 marks)

Given: specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

(ii) If the heat capacity of the container is **not** negligible, explain whether more ice, less ice or the same amount of ice is needed to obtain the final temperature of 10 °C. (2 marks)

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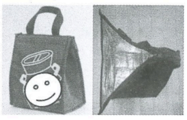
(b) Some ice cream at −10 °C is put into a 'thermal bag', of which the inner layer is made of polyethylene foam coated with aluminium foil. The bag is also equipped with a zipper at the top.



The thermal bag is then brought outdoors on a hot sunny day.

(i) Referring to the heat transfer processes, explain **ONE** feature of this bag that helps keep the ice cream at a low temperature. (1 mark)

(ii) Suggest **ONE** modification to this bag that would enhance its ability to keep things stored inside at a low temperature. (1 mark)

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Q1

1. A 150 W immersion heater is used to keep the water in a large beaker boiling under standard atmospheri pressure. In 5 minutes, 16 g of water boils away. Neglect any heat loss to surroundings.

(a) Find the specific latent heat of vaporization of water, l. (2 marks

A student puts a small metal sphere in the boiling water. After a few minutes, the sphere is quickly transferre to a polystyrene cup containing 100 g of water at a temperature of 20 °C. The cup of water is stirred gently an its highest temperature attained is 22 °C.
Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

(b) Estimate the heat capacity C of the metal sphere. (2 marks

(c) In fact the sphere has carried with it some boiling water to the cup of water. Referring to this fact, explai whether the true value of C is higher or lower than the value calculated in (b). (2 marks

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(d) In order to reduce the error contributed by the polystyrene cup, another student suggests repeating the measurements using a copper cup of similar shape and size. Explain whether the suggestion is justified. (2 marks)

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Q1

C

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81\ \text{m s}^{-2}$.

1. A bottle of water of temperature 4 °C is taken out from a refrigerator and placed on a table. The room temperature is 30 °C.

Figure 1.1

(a) State one way of heat transfer from the surroundings to the bottle of water. (1 mark)

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(b) Explain whether wrapping the bottle with an aluminum foil will speed up the rate of temperature increase. (2 marks)

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(c) After a period of time, water droplets form on the surface of the bottle.

(i) State the source of the water droplets. (1 mark)

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(ii) Suppose 0.40 g of water droplets are collected on the surface of the bottle. Estimate the latent heat released from these droplets. (2 marks)
Given: specific latent heat of vaporization of water = $2.26 \times 10^{6}\ \text{J kg}^{-1}$

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Q1

1. A mountaineer arrives at a location *X*, which is 3000 m above sea level. At *X*, the melting point of ice is 0 °C and the boiling point of water is 90 °C. The mountaineer needs to heat 0.50 kg of melting ice to its boiling point using a heater.

(a) Estimate the energy required. (2 marks)

Given: specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

specific heat capacity of water = 4200 J kg^{-1} °C^{-1}